

E-Commerce Analytics: From SQL Business Questions to Interactive Business Intelligence

1. Project Overview

This project analyzes a multi-year e-commerce database using Microsoft SQL Server and Power BI. The objective is to transform normalized transactional tables into reproducible business analysis and then convert that logic into an interactive decision-support dashboard. The database model connects customers, orders, order items, products, categories, sellers, payments, shipping, and inventory so that commercial and operational questions can be answered from a single integrated model.

The workflow was designed as a connected analytics process rather than a collection of isolated visuals. SQL was used first to validate joins, calculate revenue and order-level measures, apply ranking and window functions, identify operational exceptions, and implement an inventory trigger. Power BI then reused the same business logic through relationships, DAX measures, filters, matrices, tooltips, and page navigation. The final report contains five pages: Executive Sales Overview, Customer Analytics, Product & Inventory Analytics, Orders, Payments & Shipping, and Seller & Geographic Performance.

Project Component	Implementation
Database / Querying	Microsoft SQL Server Management Studio (SSMS)
Core Data Model	category, customers, inventory, order_items, orders, payments, products, sellers, shipping
Time Intelligence	Power BI DateTable with Year, Month, Month Start, and Year-Month fields
Reporting	Five-page interactive Power BI report with Category and Year slicers plus page-navigation buttons
Primary Analysis Themes	Sales, customers, products, inventory, payments, shipping, seller performance, geographic performance, ranking, cross-sell, and revenue decline

2. Dataset and Analytical Model Summary

The project uses a relational e-commerce model centered on the orders and order_items tables. Customers and sellers connect to orders, order_items connect orders to products, products connect to category and inventory, while payments and shipping provide order-level operational status. This structure supports both descriptive analysis and cross-functional reporting without duplicating transactional logic.

Dataset Characteristic	Value / Description
Total Revenue	\$12.64M (\$12,642,070.56)
Distinct Orders	21,629
Customers	898
Products	765

Dataset Characteristic	Value / Description
Sellers	54
Shipping Records	21,141
Product Categories	6
Average Order Value	\$584.50
Sales History	Multi-year history spanning 2020 through 2024
Core Dimensions / Entities	Customer, product, category, seller, date, payment, shipping, and inventory

The model is suitable for both business performance analysis and operational monitoring. Customer-level measures can be evaluated by state and purchase behavior, product measures by category and inventory level, seller measures by revenue and activity status, and order measures by payment and shipping condition. Power BI adds a dedicated DateTable so yearly and monthly filters can be applied consistently across report pages.

3. Exploratory Data Analysis using SQL

The SQL stage begins by answering core business questions directly against the relational tables. The analysis combines aggregation, joins, common table expressions (CTEs), ranking functions, conditional aggregation, anti-joins, date functions, and a trigger. Each query was designed to answer a specific operational or commercial question that could later be represented in Power BI.

- **Sales and Revenue Exploration:** SUM(), COUNT(), COUNT(DISTINCT), GROUP BY, and TOP were used to calculate total sales value, quantity, category revenue, seller revenue, and product rankings.
- **Customer Exploration:** Customer order frequency, average order value, lifetime spending, no-purchase customers, customer categories, cross-sell opportunities, and state-level rankings were analyzed.
- **Product and Inventory Exploration:** Product profit, returned units, return rate, low-stock alerts, and 2022-versus-2023 revenue decline were calculated.
- **Operational Exploration:** Payment-status shares, delayed shipments, paid orders still in progress, provider-level shipping performance, and inactive sellers were identified.
- **Automation:** An AFTER INSERT trigger reduces inventory automatically when new order-item records are inserted, including multi-row inserts.

Business Question	SQL Technique	Dashboard Alignment
Which products generate the most revenue?	TOP 10 + SUM(quantity * price_per_unit)	Top 10 Selling Products by Revenue
Which categories drive revenue?	CTE + GROUP BY + windowed SUM()	Revenue by Category
Which customers have the highest lifetime value?	CTE + SUM() + RANK()	Top 10 Customers by Lifetime Value
Who has not purchased?	LEFT JOIN + IS NULL anti-join	Customers With No Purchases
Which products are low in stock?	JOIN + stock < 10	Low Stock Inventory
Which orders are shipping late?	DATEDIFF / DATEADD filter	Shipping Delays (>3 Days)
Which sellers perform best?	Conditional COUNT + revenue + TOP 5	Top 5 Sellers by Revenue
Which customers are cross-sell targets?	NOT EXISTS	Cross-Sell Opportunities

Business Question	SQL Technique	Dashboard Alignment
Which products declined most year-over-year?	Conditional SUM + percentage change	Top 10 Products by Revenue Decline

4. Advanced Analytics using SQL

The advanced SQL stage moves beyond static totals and introduces time-series comparison, customer-level behavior, exception detection, ranking within groups, and business-process automation. These queries form the analytical backbone of the Power BI report.

4.1 Sales Trend and Revenue Analysis

Monthly sales are calculated from the orders and order_items tables using the first day of each month as a stable month key. The analysis uses the latest date in the dataset rather than the current system date, which makes the rolling 12-month logic reproducible for historical data. LAG() then places the previous month's sales beside the current month for direct comparison.

```
WITH current_month_sales AS (
    SELECT DATEFROMPARTS(YEAR(o.order_date), MONTH(o.order_date), 1) AS sales_month,
           SUM(oi.quantity * oi.price_per_unit) AS sales_amount_per_month
    FROM orders o
    LEFT JOIN order_items oi ON o.order_id = oi.order_id
    GROUP BY DATEFROMPARTS(YEAR(o.order_date), MONTH(o.order_date), 1)
)
SELECT sales_month,
       sales_amount_per_month,
       LAG(sales_amount_per_month) OVER (ORDER BY sales_month) AS previous_month_sales
FROM current_month_sales;
```

Category contribution extends this logic by aggregating product revenue through the product-category relationship and dividing each category by overall revenue. The dashboard shows that electronics is overwhelmingly the dominant revenue category, accounting for roughly nine-tenths of total revenue.

4.2 Customer and Product Intelligence

Customer analysis uses an order-level CTE to avoid averaging line items when calculating average order value. Lifetime value is calculated by summing all historical purchases and ranking customers from highest to lowest. Customers with no purchases are found with a LEFT JOIN anti-join, while state-level customer rankings use RANK() with PARTITION BY so customers compete only with others in the same state.

The cross-sell query uses NOT EXISTS to identify customers who bought Apple AirPods 3rd Gen but never bought Apple AirPods Max. This creates a direct business-use case from transaction history: identify a known buyer of a related product and recommend the missing complementary item.

```
WHERE p.product_name = 'Apple AirPods 3rd Gen'
AND NOT EXISTS (
    SELECT 1
    FROM orders o2
    JOIN order_items oi2 ON o2.order_id = oi2.order_id
    JOIN products p2 ON p2.product_id = oi2.product_id
    WHERE o2.customer_id = c.Customer_ID
    AND p2.product_name = 'Apple AirPods Max'
);
```

Product analytics combines profitability, returns, and year-over-year decline. Unit profit is calculated as price minus cost of goods sold (COGS). Return rate divides returned units by total units sold using NULLIF() to protect the denominator. Revenue-decline analysis compares 2022 and 2023 sales and isolates products whose revenue decreased.

4.3 Operational Monitoring and Automation

Operational queries connect revenue performance with fulfillment and payment behavior. Delayed orders are defined as shipments occurring more than three days after the order date. Paid orders still marked Inprogress are surfaced as pending shipment exceptions. Shipping providers are compared by revenue, order count, and average shipping days, while seller activity is evaluated relative to the latest order date in the dataset.

The inventory trigger automates stock reduction whenever new sales records are inserted into order_items. The inserted pseudo-table is grouped by product so the trigger remains correct even when multiple rows are inserted in a single statement.

```
CREATE TRIGGER trg_reduce_inventory
ON order_items
AFTER INSERT
AS
BEGIN
    UPDATE i
    SET i.stock = i.stock - x.quantity_sold
    FROM inventory i
    JOIN (
        SELECT product_id, SUM(quantity) AS quantity_sold
        FROM inserted
        GROUP BY product_id
    ) x ON i.product_id = x.product_id;
END;
```

5. SQL Refinements and Enhancements

The source SQL provides the analytical foundation for the dashboard. During report development, several refinements were applied or identified to make the analysis more reliable, reproducible, and Power BI-friendly. These refinements preserve the intent of the original business questions while improving implementation quality.

Area	Refinement / Enhancement	Why It Improves the Analysis
Customer Identity	Use Customer_ID or a Customer Label rather than customer name alone in BI visuals.	Prevents different people with identical names from being merged into one customer.
Monthly Analysis	Use DATEFROMPARTS / Month Start and dataset MAX(order_date).	Keeps months separate across years and makes historical windows reproducible.
Average Order Value	Aggregate each order first, then average order values by customer.	Prevents line-item counts from distorting customer AOV.
Top-N Ties	Use deterministic tie-breakers for exact Top 10 outputs when many products share the same decline.	Makes tied results reproducible rather than dependent on arbitrary row order.
Return Rate	Use NULLIF(total_units, 0) in SQL and DIVIDE() in DAX.	Avoids divide-by-zero errors.

Area	Refinement / Enhancement	Why It Improves the Analysis
Inactive Sellers	Use LEFT JOIN, COALESCE(), and the dataset's latest order date.	Includes sellers with no orders and keeps inactivity logic tied to the data period.
Shipping Provider Logic	Relate shipping records to orders explicitly by order_id before revenue aggregation.	Keeps provider revenue and order counts at the correct grain.
Inventory Trigger	Aggregate inserted rows by product before stock update.	Makes the trigger safe for multi-row inserts.

6. Dashboard in Power BI

The Power BI report converts the SQL analysis into a business-facing interface. A consistent burgundy-and-rose visual theme is used across all five pages. Category and Year slicers allow the analytical context to change without rewriting SQL, while custom navigation buttons provide direct movement between pages. DAX measures reproduce the SQL logic for revenue, orders, average order value, returns, shipping, seller activity, and ranking.

6.1 Executive Sales Overview

The Executive Sales Overview summarizes overall commercial performance. KPI cards show \$12.64M in revenue, approximately 22K orders, 898 customers, and a \$584.50 average order value. The page combines the monthly sales trend, revenue by category, the top ten selling products by revenue, and payment-status composition. Payment Succeeded accounts for 84.61% of payment records, Refunded for 13.13%, and Payment Failed for 2.26%.

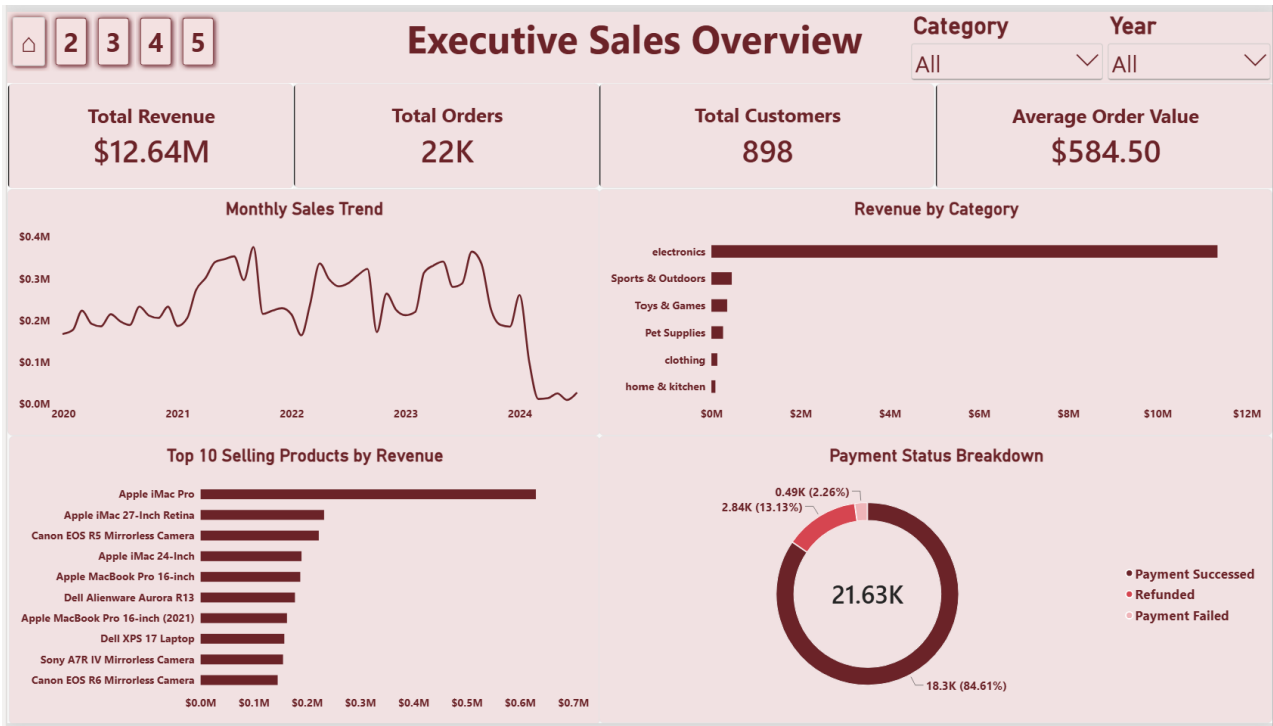


Figure 1. Executive Sales Overview - overall revenue, sales trend, category mix, product ranking, and payment status.

6.2 Customer Analytics

The Customer Analytics page translates customer-focused SQL questions into rankings, segmentation, and exception tables. The dataset contains 898 customers, including 212 with no purchases. Under the project rule, a

customer with more than five returned orders is classified as Returning; 30 customers meet that rule while 868 remain New. The page also shows the top ten customers by lifetime value, average order value for customers with more than five orders, cross-sell opportunities, and top customers by orders within each state.

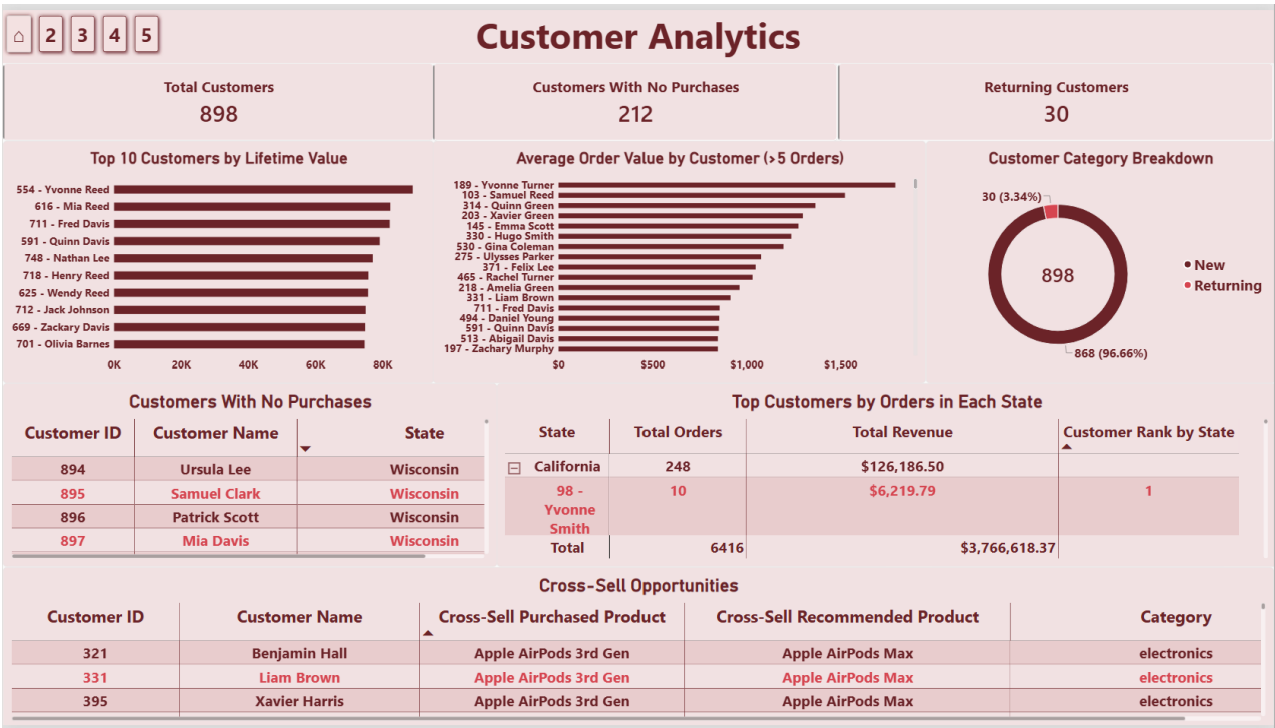


Figure 2. Customer Analytics - lifetime value, average order value, customer category, no-purchase customers, state ranking, and cross-sell opportunities.

6.3 Product & Inventory Analytics

The Product & Inventory Analytics page combines profitability, return behavior, inventory risk, and year-over-year decline. The dataset contains 765 products, with 51 products currently below 10 units of stock. Returned units total approximately 5K. Product profit is defined as selling price minus COGS, while the return-rate measure compares returned units with total units sold. The revenue-decline table identifies ten products whose 2023 revenue fell completely to zero from a positive 2022 base, producing a -100% decline.

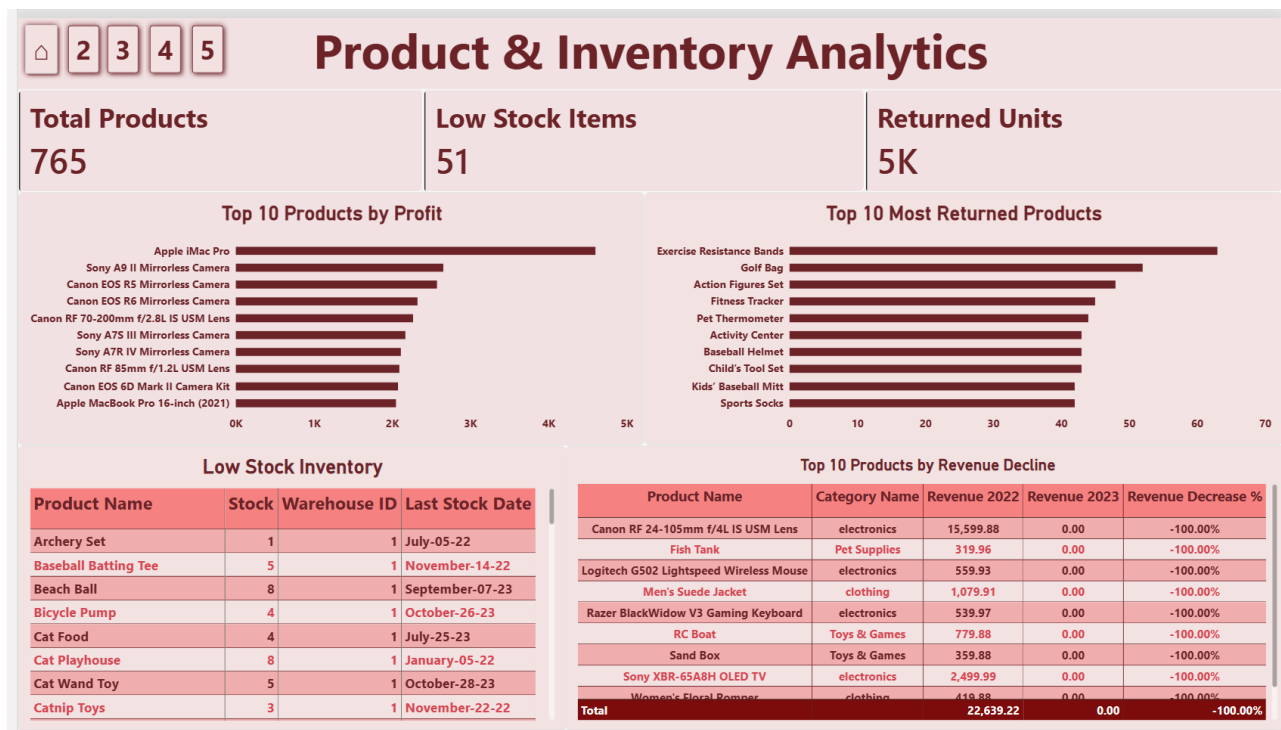


Figure 3. Product & Inventory Analytics - product profit, returned products, low-stock inventory, and 2022-to-2023 revenue decline.

6.4 Orders, Payments & Shipping

The Orders, Payments & Shipping page focuses on fulfillment and transaction quality. Payment success is 84.61%, approximately 8K orders exceed the three-day shipping-delay threshold, and 499 orders are successfully paid but still marked Inprogress. Revenue is compared across FedEx, DHL, and Bluedart, with average shipping days and provider order counts available through tooltips. Two detailed tables expose delayed shipments and paid orders pending shipment for operational follow-up.



Figure 4. Orders, Payments & Shipping - payment success, shipping-provider revenue, delay exceptions, and pending shipments.

6.5 Seller & Geographic Performance

The Seller & Geographic Performance page evaluates seller revenue, seller activity, geographic category performance, and seller origin. Of 54 sellers, 52 are active and 2 are inactive. The inactive sellers are Clorox and Lysol, both with no recorded last order and zero revenue. The leading sellers by revenue are AnkerDirect, Tech Armor, AmazonBasics, iSaddle, and Ailun. Seller origin is led by the USA with 23 sellers (42.59%), followed by China with 15 (27.78%), Canada with 9 (16.67%), and the UK with 7 (12.96%).

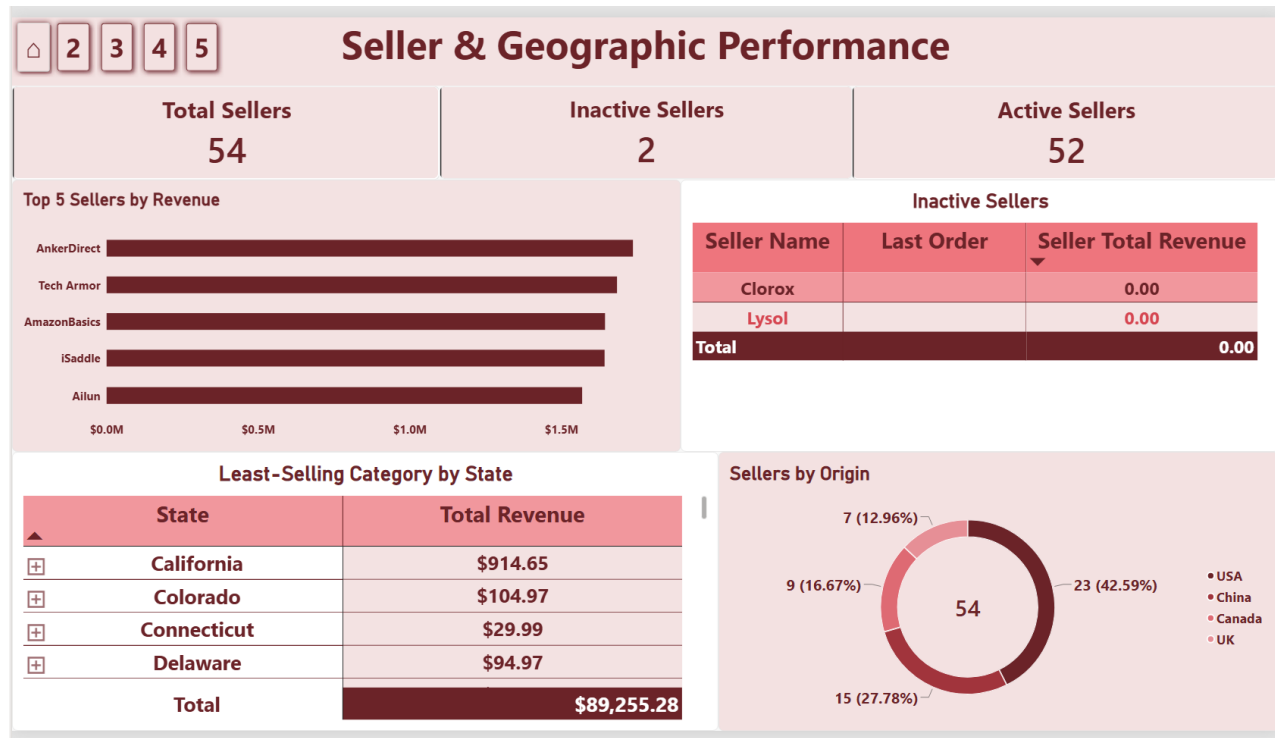


Figure 5. Seller & Geographic Performance - seller KPIs, top sellers, inactive sellers, least-selling category by state, and seller origin.

7. Key Findings and Interpretations

- **Revenue is highly concentrated by category:** Electronics contributes the overwhelming majority of revenue, so overall performance is strongly tied to electronics demand.
- **Payment performance is strong but not perfect:** 84.61% of payment records are successful, while refunds account for 13.13% and failures for 2.26%.
- **A meaningful share of customers have no purchase history:** 212 of 898 customers (about 23.6%) have no orders, creating a clear activation opportunity.
- **Customer returns are concentrated in a small group:** Only 30 customers (3.34%) meet the project's Returning definition of more than five returned orders.
- **Inventory risk is visible but contained:** 51 of 765 products (about 6.7%) are below 10 units of stock and should be prioritized for replenishment review.
- **Year-over-year product decline can be severe:** The ten products shown in the decline table lost 100% of their 2022 revenue in 2023, indicating discontinued demand, stock availability issues, or product lifecycle changes that require investigation.
- **Fulfillment is a major operational issue:** Approximately 8K orders exceed the three-day shipping threshold, and 499 successfully paid orders remain Inprogress.
- **Seller activity is broadly healthy:** 52 of 54 sellers are active; only Clorox and Lysol are inactive under the six-month rule.
- **Seller geography is diversified:** The USA supplies 42.59% of sellers, with the remaining seller base distributed across China, Canada, and the UK.

8. Business Recommendations

- **Reduce category concentration risk:** Protect high-performing electronics products while testing growth opportunities in Sports & Outdoors, Toys & Games, Pet Supplies, Clothing, and Home & Kitchen.
- **Activate no-purchase customers:** Use onboarding campaigns, targeted first-purchase incentives, and category recommendations for the 212 customers with no recorded orders.
- **Use cross-sell opportunities operationally:** Customers who purchased Apple AirPods 3rd Gen but not Apple AirPods Max can be targeted with related-product recommendations or bundles.
- **Prioritize low-stock replenishment:** Review the 51 low-stock products by revenue importance, return rate, and warehouse location before replenishment decisions are made.
- **Investigate high-return products and customers:** Combine the Top Returned Products view with customer-return behavior to identify product-quality, expectation, or fulfillment issues.
- **Improve shipping performance by provider:** Use provider revenue, average shipping days, and delayed-order detail together to identify whether delay problems are concentrated with specific providers or time periods.
- **Resolve paid-but-pending orders quickly:** The 499 successfully paid Inprogress orders should be monitored as an exception queue because they directly affect customer experience and working capital.
- **Review products with complete revenue loss:** Products showing -100% revenue decline from 2022 to 2023 should be checked for discontinuation, stockouts, replacement models, or demand shifts before strategic decisions are made.
- **Retain strong sellers and reactivate inactive sellers selectively:** Top sellers should receive appropriate inventory and operational support, while inactive sellers should be reviewed for reactivation potential or rationalization.

9. Conclusion and Future Scope

This project demonstrates a complete SQL-to-BI analytics workflow. SQL establishes the database relationships and answers 21 business questions covering revenue, customers, products, inventory, payments, shipping, sellers, ranking, cross-sell, and automation. Power BI converts those results into an interactive five-page report that preserves the business logic while making the findings accessible through filters, tooltips, matrices, KPI cards, and navigation.

The strongest technical value of the project is the connection between query-level logic and dashboard-level decision support. The Executive Sales Overview summarizes the business; Customer Analytics explains value, inactivity, returns, ranking, and cross-sell; Product & Inventory Analytics links profitability with stock and returns; Orders, Payments & Shipping exposes transaction and fulfillment exceptions; and Seller & Geographic Performance combines revenue, activity, state-level category performance, and seller origin.

Future work can extend the model with gross-margin analysis at transaction level, customer cohorts, churn and repeat-purchase prediction, demand forecasting, supplier or warehouse performance, delivery completion dates, stockout forecasting, and automated data refresh. A production version could also move more reusable calculations into curated SQL views, add row-level security in Power BI, introduce drill-through pages for customer and product detail, and schedule alerts for low stock, payment failures, shipping delays, and inactive sellers.